BioMind-BG: A Transparent, Biologically Grounded Spiking Model of the Cortico-Basal-Ganglia-Thalamic Circuit with Interpretable Dopamine-Driven Plasticity

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Abstract

Existing spiking models of the cortico-basal-ganglia-thalamic (CBGT) circuit, while biologically detailed, are often embedded in complex software frameworks that obscure the mapping between biological mechanisms and computational elements. This obscurity limits interpretability, reproducibility, and integration with other brain-inspired modules. We present BioMind-BG, an open-source spiking neural network model of the CBGT circuit in which every computational element—neuron parameters, synaptic pathways, receptor kinetics, and plasticity rules—maps directly and traceably to identified biological mechanisms. The model simulates nine neural populations (∼3,000 leaky integrate-and-fire neurons) connected by 27 biologically constrained pathways across three receptor types (AMPA, $GABA_A$, NMDA), with a three-factor corticostriatal learning rule combining spike-timing-dependent plasticity, eligibility traces, and D1/D2-specific dopaminergic modulation. Implemented in ∼900 lines of pure Python/NumPy with no external neural simulation frameworks, the model achieves three objectives: (1) all nine populations produce spontaneous firing rates within established electrophysiological ranges without parameter fitting; (2) lesion manipulations accurately reproduce the clinical phenotypes of Parkinson's disease (elevated GPi, suppressed thalamus), Huntington's disease (reduced GPi, elevated thalamus), and subthalamic deep brain stimulation; and (3) the architecture is designed for modular integration with additional brain-inspired components. We further demonstrate reward-driven action selection learning and characterize a biologically realistic perseveration effect during reward reversal that reflects the known dissociation between goal-directed and habitual control systems. All reported results are consistent across five independent runs with different random seeds. Code, documentation, and experimental scripts are publicly available at:

`https://github.com/Rekhii/Biomind`

Keywords: basal ganglia, spiking neural network, action selection, dopamine, reward learning, corticostriatal plasticity, computational neuroscience, interpretable neural modeling

1 Introduction

The basal ganglia (BG) implement a competitive selection process in which striatal medium spiny neurons (MSNs) selectively inhibit tonically active output nuclei, thereby releasing thalamic relay neurons from inhibition and allowing selected motor programs, cognitive strategies, or motivational states to access cortical processing resources (Mink, 1996; Redgrave et al., 1999; Gurney et al., 2001). The clinical significance of this circuit is well established: Parkinson's disease produces bradykinesia through excessive output nucleus activity (DeLong, 1990), while Huntington's disease produces chorea through insufficient output nucleus activity (Albin et al.,

1989). These disease phenotypes provide powerful, clinically grounded validation criteria for computational models.

Several spiking models of the CBGT circuit have been developed, including the influential work of Humphries et al. (2006) and the CBGTPy framework of Dunovan et al. (2019). These models have advanced our understanding of BG dynamics in decision-making. However, they typically embed biological mechanisms within complex software infrastructure—custom data structures, tracing systems, parallelization layers, and DataFrame-based parameter management—that makes it difficult for a reader to trace any given computational operation back to its biological origin.

This paper introduces BioMind-BG, a reimplementation of the CBGT spiking circuit that prioritizes *interpretability* and *traceability*. Our contribution is not a novel circuit topology or a new plasticity rule; rather, it is a model in which the mapping between biology and computation is explicit at every level:

- Every neuron parameter is collected in a single documented file with biological units and literature sources, rather than distributed across multiple configuration layers.
- Every synaptic pathway is defined as a single human-readable tuple (source, destination, receptor, type, probability, efficacy, plasticity), rather than constructed through programmatic grid-filling operations.
- Every differential equation in the simulation loop is annotated with its biological identity (leak current, T-type calcium rebound, NMDA Mg^{2+} block, dopaminergic eligibility trace), rather than appearing as undifferentiated matrix operations.

This transparency serves two purposes. First, it enables researchers to inspect, modify, and extend any biological mechanism without navigating abstraction layers. Second, it facilitates integration with additional brain-inspired modules—the model is designed as the first component of a multi-region architecture, with clearly defined input/output interfaces at the cortical and thalamic boundaries.

We validate BioMind-BG across five experiments: baseline electrophysiology, reward-driven learning, reward reversal, lesion studies reproducing clinical phenotypes, and multi-choice scaling analysis. All results are consistent across repeated runs with different random seeds, and the model achieves biologically plausible dynamics without post-hoc parameter fitting—all parameters are drawn directly from the biophysical literature or from the validated CBGTPy framework.

2 Biological Background

This section summarizes the biological mechanisms that the model implements. Each mechanism described here corresponds directly to a computational element in Section 3; we note these correspondences explicitly to maintain the traceability that is central to our approach.

2.1 Circuit Architecture

The CBGT circuit comprises nine functionally distinct neural populations organized into three canonical pathways (Alexander et al., 1986):

Direct pathway (Go): Cortex $\xrightarrow{\text{Glu}}$ D1-MSNs (dSPN) $\xrightarrow{\text{GABA}}$ GPi $\dashv$ Thalamus. Activation of D1-MSNs inhibits GPi, disinhibiting thalamus and facilitating the selected action. *In the model:* this corresponds to Cx→dSPN AMPA/NMDA excitation (Eq. 4), dSPN→GPi GABA inhibition, and GPi→Th GABA inhibition, all defined as pathway tuples in Table 2.

Indirect pathway (NoGo): Cortex $\xrightarrow{\text{Glu}}$ D2-MSNs (iSPN) $\xrightarrow{\text{GABA}}$ GPe $\xrightarrow{\text{GABA}}$ STN $\xrightarrow{\text{Glu}}$ GPi $\rightarrow$ Thalamus. This multi-synaptic pathway increases GPi output, suppressing competing actions. *In the model:* the STN→GPi projection uses the "all-to-all" connection type, broadcasting excitation across all action channels rather than selectively (Table 2), implementing the broad "brake" function proposed by Nambu et al. (2002).

Additional circuit elements include fast-spiking interneurons (FSI) providing feedforward inhibition to MSNs, lateral inhibition among MSNs within each action channel, cortical interneurons (CxI) mediating local E-I balance, and thalamocortical and thalamostriatal feedback loops that close the re-entrant circuit.

2.2 Dopaminergic Modulation

Dopamine from the substantia nigra pars compacta (SNc) differentially modulates the two striatal populations through D1 and D2 receptor subtypes (Gerfen & Surmeier, 2011). D1 activation enhances dSPN excitability; D2 activation reduces iSPN excitability. Phasic dopamine bursts (positive reward prediction error) simultaneously strengthen the Go pathway and weaken the NoGo pathway, while dopamine dips (negative prediction error) produce the opposite effect (Frank, 2004).

The learning rule at corticostriatal synapses requires three factors: (1) a spike-timing-dependent plasticity (STDP) signal, (2) an eligibility trace bridging the temporal gap between spike correlations ($\sim$20 ms) and dopamine arrival ($\sim$100–500 ms), and (3) the dopamine signal itself, transformed through receptor-specific nonlinearities (Izhikevich, 2007; Yagishita et al., 2014). *In the model:* these three factors correspond to Equations 7–9, 10, and 11, respectively.

3 Model Description

3.1 Neuron Model

Each neuron is modeled as a leaky integrate-and-fire (LIF) unit with extensions for calcium-dependent afterhyperpolarization and voltage-gated conductances:

$$\frac{dV}{dt} = -\frac{V - V_{\text{rest}}}{\tau_m} - \frac{g_{\text{Ca}} \cdot g_{\text{AHP}}}{C}(V - V_K) - \frac{g_{\text{ADR}}}{C}(V - V_{\text{ADR}}) - \frac{g_K}{C}(V - V_{\text{ADR}}) - \frac{g_{\text{rb}}}{C}(V - V_T) \quad (1)$$

where $V_{\text{rest}} = -70$ mV is the resting potential, τ_m is the membrane time constant (population-specific; see Table 1), C is the membrane capacitance, and $V_\theta = -50$ mV is the spike threshold. Upon threshold crossing, V is reset to $V_{\text{reset}} = -55$ mV and the neuron enters a 2 ms refractory period. The five terms represent, respectively: passive leak toward resting potential, calcium-activated potassium current (spike-rate adaptation), anomalous delayed rectification, outward rectifying potassium current, and low-threshold calcium (T-type) rebound current. The last three conductances are set to zero for most populations; their nonzero assignments are listed in Table 1.

The T-type calcium current ($g_{\text{rb}} = g_T \cdot h$) enables rebound bursting in STN and GPe ($g_T = 0.06$ mS/cm^2). The gating variable h de-inactivates during hyperpolarization and inactivates during depolarization:

$$\frac{dh}{dt} = \begin{cases} (1-h)/\tau_{hp} & \text{if } V < V_h \\ -h/\tau_{hm} & \text{if } V \geq V_h \end{cases} \quad (2)$$

with $V_h = -60$ mV, $\tau_{hp} = 100$ ms (de-inactivation), and $\tau_{hm} = 20$ ms (inactivation). This mechanism allows STN neurons to fire rebound bursts following release from GPe inhibition, a well-characterized electrophysiological property (Bevan et al., 2002).

Table 1: Population-specific parameters. All other parameters use the defaults: $C = 0.5$ nF, $\tau_m = 20$ ms, $g_T = 0$. The column N gives neurons per action channel.

Population	Role	N	C (nF)	τ_m (ms)	g_T
Cx (Cortex)	Input encoding	204	0.5	20	0
CxI (Cx interneurons)	E-I balance	186	0.2	10	0
dSPN (D1-MSN)	Go pathway	75	0.5	20	0
iSPN (D2-MSN)	NoGo pathway	75	0.5	20	0
FSI	Striatal gain control	75	0.2	10	0
GPe	Indirect relay	750	0.5	20	0.06
STN	Excitatory brake	750	0.5	20	0.06
GPi	BG output gate	75	0.5	20	0
Th (Thalamus)	Decision relay	75	0.5	27.78	0

3.2 Synaptic Transmission

Three receptor types mediate synaptic transmission, each with distinct kinetics. AMPA ($\tau = 2$ ms, $E_{\text{rev}} = 0$ mV) provides fast excitation; GABA_A ($\tau = 5$ ms, $E_{\text{rev}} = -70$ mV) provides fast inhibition; NMDA ($\tau = 100$ ms, $E_{\text{rev}} = 0$ mV) provides slow, sustained excitation with voltage-dependent Mg^{2+} block.

External background input models the aggregate effect of unmodeled afferent connections as an Ornstein-Uhlenbeck process:

$$dS_R^{\text{ext}} = \frac{dt}{\tau_R}(-S_R^{\text{ext}} + \mu_R) + \sigma_R\sqrt{\frac{2\,dt}{\tau_R}}\,\mathcal{N}(0,1) \tag{3}$$

where $\mu_R = \eta_R \cdot f_R \cdot N_R \cdot \tau_R$ and $\sigma_R = \eta_R\sqrt{\tau_R \cdot f_R \cdot N_R/2}$, with η_R the synaptic efficacy, f_R the input frequency, and N_R the number of external connections.

Lateral (network-driven) conductances decay exponentially and receive discrete increments upon presynaptic spikes:

$$S_R^{\text{lat}}(t) = S_R^{\text{lat}}(t^-) \cdot e^{-dt/\tau_R} + \sum_{j \in \text{spikes}} w_{ij} \cdot c_{ij} \tag{4}$$

where w_{ij} is the synaptic efficacy and $c_{ij} \in \{0, 1\}$ indicates the existence of a connection.

The NMDA current includes voltage-dependent Mg^{2+} block (Jahr & Stevens, 1990):

$$I_{\text{NMDA}} = \frac{(E_{\text{NMDA}} - V) \cdot S_{\text{NMDA}}}{1 + \exp(-0.062 \cdot V/3.57)} \tag{5}$$

with a saturation mechanism ($\alpha = 0.6332$) that prevents unbounded conductance accumulation during sustained presynaptic firing.

The complete synaptic current for each neuron sums external and lateral contributions across all three receptor types:

$$I_{\text{syn}} = \sum_{R \in \{\text{AMPA, GABA, NMDA}\}} \frac{(E_R - V) \cdot (S_R^{\text{lat}} + S_R^{\text{ext}})}{C} \cdot B_R(V) \tag{6}$$

where $B_R(V) = 1$ for AMPA and GABA, and $B_R(V) = [1 + \exp(-0.062V/3.57)]^{-1}$ for NMDA.

3.3 Pathway Connectivity

All 27 synaptic projections are defined as tuples specifying source population, destination population, receptor type, connection topology, connection probability, synaptic efficacy, and plasticity status (Table 2). The connection topology field distinguishes channel-specific ("syn")

Table 2: Selected synaptic pathways. "syn" = within-channel; "all" = across all channels. Only Cx→dSPN and Cx→iSPN AMPA connections are plastic. Full pathway table: 34 projections total.

Source	Dest	Receptor	Type	Prob	Eff	Notes
Direct pathway (Go)						
Cx	dSPN	AMPA	syn	1.0	0.015	Plastic
Cx	dSPN	NMDA	syn	1.0	0.02	
dSPN	GPi	GABA	syn	1.0	2.09	Disinhibition gate
Indirect pathway (NoGo)						
Cx	iSPN	AMPA	syn	1.0	0.015	Plastic
iSPN	GPe	GABA	syn	1.0	4.07	
GPe	STN	GABA	syn	0.067	0.35	Sparse inhibition
STN	GPi	NMDA	all	1.0	0.038	Broad brake
Output and feedback						
GPi	Th	GABA	syn	1.0	0.33	BG output gate
Th	Cx	NMDA	all	0.83	0.03	Thalamocortical
Th	dSPN	AMPA	syn	1.0	0.38	Thalamostriatal

projections, which connect only within the same action channel, from all-to-all ("all") projections, which connect across channels.

3.4 Three-Factor Learning Rule

Corticostriatal plasticity operates only at Cx→dSPN and Cx→iSPN AMPA synapses, consistent with the biological locus of dopamine-dependent learning (Yagishita et al., 2014). The rule combines three factors:

Factor 1: STDP traces. Pre- and post-synaptic activity traces are updated as:

$$\frac{dA_{\text{pre}}}{dt} = \frac{\delta_{\text{pre}} \cdot X_{\text{pre}} - A_{\text{pre}}}{\tau_{\text{pre}}} \tag{7}$$

$$\frac{dA_{\text{post}}}{dt} = \frac{\delta_{\text{post}} \cdot X_{\text{post}} - A_{\text{post}}}{\tau_{\text{post}}} \tag{8}$$

where $X_{\text{pre/post}} \in \{0, 1\}$ are spike indicators, $\delta_{\text{pre}} = 0.8$, $\delta_{\text{post}} = 0.04$, $\tau_{\text{pre}} = 15$ ms, $\tau_{\text{post}} = 6$ ms. The asymmetry ($\delta_{\text{pre}} \gg \delta_{\text{post}}$) ensures that cortical input timing dominates the eligibility trace.

Factor 2: Eligibility trace. The traces combine into a direction signal:

$$\frac{dE}{dt} = \frac{X_{\text{post}} \cdot A_{\text{pre}} - X_{\text{pre}} \cdot A_{\text{post}} - E}{\tau_E} \tag{9}$$

with $\tau_E = 100$ ms. Positive E indicates pre-before-post timing (LTP direction); negative E indicates post-before-pre (LTD direction). The 100 ms decay bridges the temporal gap to dopamine arrival.

Factor 3: Dopamine signal. The phasic dopamine signal is transformed through receptor-specific nonlinearities:

$$f_{\text{D1}}(\text{DA}) = \begin{cases} (y/x) \cdot \text{DA} & \text{if DA} > -x \\ -y & \text{otherwise} \end{cases} \qquad f_{\text{D2}}(\text{DA}) = \begin{cases} (y/x) \cdot \text{DA} \cdot \varepsilon & \text{if DA} < x \\ y \cdot \varepsilon & \text{otherwise} \end{cases} \tag{10}$$

with $x = 0.5$, $y = 3.0$, $\varepsilon = 0.3$. The ε parameter implements the lower maximal response of D2 relative to D1 receptors, consistent with receptor pharmacology (Gerfen & Surmeier, 2011). The saturation prevents catastrophic weight changes during large prediction errors.

The weight update uses soft multiplicative bounds:

$$\Delta w = dt \cdot c_{ij} \cdot \alpha_w \cdot f(\mathrm{DA}) \cdot E \cdot \begin{cases} (w_{\max} - w) & \text{if } \Delta w > 0 \\ (w - w_{\min}) & \text{if } \Delta w < 0 \end{cases} \tag{11}$$

where $\alpha_w = +39.5$ for D1 (LTP on positive DA) and $\alpha_w = -38.2$ for D2 (LTD on positive DA), implementing the opponent Go/NoGo learning directions. The sign of α_w is the computational mechanism that implements the differential effect of dopamine on the two striatal populations.

3.5 Decision Mechanism and Reward Processing

Decisions are detected by monitoring thalamic firing rates over a 60 ms sliding window. The first action channel to exceed 30 Hz is selected (first-past-the-post competition). If no channel crosses threshold within 1000 ms, the trial is classified as a timeout.

Following a 300 ms post-decision delay, reward is delivered and the dopamine signal is computed as:

$$\mathrm{DA}_{\mathrm{phasic}} = (r - Q_a) \cdot C_{\mathrm{scale}} \tag{12}$$

where r is the reward, Q_a is the Q-value for the chosen action (updated via $Q_a \leftarrow Q_a + \alpha_Q(r - Q_a)$ with $\alpha_Q = 0.1$), and $C_{\mathrm{scale}} = 80$ converts the abstract prediction error to neural scale. This implements the reward prediction error (RPE) theory of dopamine (Schultz et al., 1997).

4 Implementation

BioMind-BG is implemented in approximately 900 lines of Python/NumPy across seven modules (Table 3). No external neural simulation frameworks (Nengo, Brian, NEST, ANNarchy) are used. The integration timestep is $dt = 0.2$ ms (forward Euler). For two action channels, the model comprises 16 populations totaling ∼3,000 neurons and 72 active synaptic projections.

Table 3: BioMind-BG module structure. Each module has a single, well-defined responsibility.

Module	Lines	Responsibility
`params.py`	190	All biological constants (single source of truth)
`populations.py`	130	Population instantiation and connectivity matrices
`agent.py`	160	State variable initialization (per-neuron arrays)
`timestep.py`	260	Simulation loop (Eqs. 1–11)
`qlearning.py`	65	Q-value tracking and RPE computation (Eq. 12)
`trial.py`	130	Three-phase trial state machine
`run.py`	170	Orchestration and result collection

Reproducibility. All experiments use fixed random seeds. Results reported here are consistent across five independent runs with seeds 42, 43, 44, 45, and 46; we report seed 42 throughout. The model runs on a standard laptop (no GPU required) at approximately 10 seconds per simulated trial (1–2 seconds of simulated time per trial including warmup, stimulus, decision, and inter-trial interval).

Design for integration. The model exposes clear interfaces at its cortical input (stimulus frequency modulation) and thalamic output (firing rate readout), enabling future connection to upstream sensory processing and downstream motor execution modules.

5 Experiments and Results

5.1 Experiment 1: Baseline Firing Rates

We validated that the network produces biologically plausible spontaneous firing rates in the absence of external task-related stimulation. After a 1000 ms warmup period (5000 integration steps), we measured average firing rates across all populations (Table 4).

Table 4: Baseline firing rates compared to primate and rodent electrophysiological recordings. All nine populations fall within established ranges without parameter fitting.

Population	Model (Hz)	Literature (Hz)	Source
Cortex (Cx)	0.0	0–5	Constantinidis & Steinmetz (2002)
Cx Interneurons (CxI)	0.8	0–10	Markram et al. (2004)
D1-MSN (dSPN)	3.8	0.5–8	Berke et al. (2004)
D2-MSN (iSPN)	4.2	0.5–8	Berke et al. (2004)
FSI	7.8	2–30	Berke et al. (2004)
GPe	60.7	20–80	DeLong (1971)
STN	24.9	8–40	Wichmann & DeLong (1994)
GPi	71.0	30–100	DeLong (1971)
Thalamus (Th)	6.8	0–20	Sherman & Guillery (2001)

Takeaway. The tonic GPi rate of 71.0 Hz maintains thalamic suppression at 6.8 Hz, consistent with the gating function of the BG output. The model achieves these rates from biophysical parameters alone, without fitting to match target firing rates.

5.2 Experiment 2: Two-Choice Reward Learning

We presented the network with a two-choice probabilistic reward task (20 trials; action 0: 80% reward probability; action 1: 20%; reward $r \in \{-1, +1\}$).

The network selected action 0 on 19/20 trials (95%), with choice proportion increasing across trial blocks: 80% in trials 1–5 and 100% in trials 6–20. The Q-value for action 0 converged toward the true reward probability ($Q_0 = 0.571$ after 20 trials) while action 1 remained at its initial value ($Q_1 = 0.350$, selected only once). Total accumulated reward was +8.

Takeaway. The three-factor dopamine learning rule successfully drives behavioral adaptation: the network learns to preferentially select the higher-reward action through corticostriatal weight modification.

5.3 Experiment 3: Reward Reversal

To assess behavioral flexibility, we reversed reward contingencies after 10 trials (Phase 1: 80/20; Phase 2: 20/80). During Phase 1, the network preferred action 0 (90% choice rate). During Phase 2, despite Q-value evidence favoring action 1 ($Q_0 = -0.267$, $Q_1 = 0.350$), the network perseverated on action 0 (100% choice rate).

This perseveration is consistent with established behavioral findings. Rodent reversal learning studies demonstrate that striatal-dependent habitual responses resist extinction over short timescales (Ragozzino, 2002). The dissociation between the Q-value system (which correctly tracks the contingency change) and the weight-based thalamic competition (which preserves the prior bias) mirrors the neurobiological distinction between goal-directed (prefrontal/dorsomedial striatal) and habitual (dorsolateral striatal) control systems (Yin & Knowlton, 2004; Daw et al., 2005).

Takeaway. The model's perseveration is a biologically realistic property of isolated BG circuits. Flexible reversal would require additional prefrontal control mechanisms not included in the current model (see Section 6.3).

5.4 Experiment 4: Lesion Studies

We simulated four pathological conditions by scaling synaptic efficacies of specific populations (Table 5).

Table 5: Lesion study results. Values are baseline firing rates (Hz) after 1000 ms warmup.

Condition	GPi	Th	dSPN	iSPN	STN
Healthy	71.0	6.8	3.8	4.2	24.9
Parkinson (D1 loss, 80%)	81.0	3.4	3.2	3.7	24.8
Parkinson (D2 overactive, 5×)	85.4	2.2	2.3	2.2	27.6
Huntington (iSPN loss, 90%)	50.4	15.0	7.1	7.1	23.6
STN lesion (90%)	0.0	46.2	18.0	17.8	37.3

Parkinson's disease. Both Parkinsonian manipulations produce elevated GPi output and suppressed thalamic activity, consistent with the rate model of DeLong (1990) and the clinical presentation of bradykinesia.

Huntington's disease. Reducing iSPN output by 90% decreases GPi firing and elevates thalamic output, modeling the hyperkinetic symptoms of early Huntington's disease (Albin et al., 1989).

STN lesion. Reducing STN output eliminates GPi tonic firing, fully releasing thalamus. This models the therapeutic mechanism of subthalamic deep brain stimulation (Limousin et al., 1998) and confirms that STN excitatory drive is essential for maintaining GPi tonic activity in this model.

Takeaway. The model correctly reproduces the qualitative direction of firing rate changes in all four conditions, matching the predictions of the classical rate model without condition-specific parameter adjustment.

5.5 Experiment 5: Multi-Choice Scaling

We tested the model with 2, 3, and 4 action channels (Table 6).

Table 6: Multi-choice scaling without input normalization.

Actions	Populations	GPi (Hz)	Th (Hz)	GPe (Hz)
2	16	71.0	6.8	60.7
3	23	161.0	0.0	49.6
4	30	218.1	0.0	41.5

Without normalization, additional channels contribute unchecked excitatory drive to GPi through the STN→GPi "all-to-all" projection, causing GPi rates to scale supralinearly and completely suppressing thalamic output for $N > 2$.

Takeaway. This result identifies a concrete scaling limitation and motivates future work on biologically grounded normalization mechanisms such as homeostatic synaptic scaling or population-level gain control.

6 Discussion

6.1 Summary of Contributions

BioMind-BG demonstrates that the core computational principles of the basal ganglia can be captured in a transparent spiking model of approximately 900 lines of code. These principles—

disinhibitory action selection, opponent Go/NoGo learning, and dopaminergic reward prediction error—are well established biologically. The primary contribution of this work is the explicit mapping between biology and computation that enables researchers to verify, modify, and extend any component without navigating abstraction layers.

6.2 Relationship to Existing Models

BioMind-BG builds upon the CBGTPy framework (Dunovan et al., 2019), using the same neuron model, receptor kinetics, and pathway parameters. It differs in three respects: (1) all parameters are consolidated in a single file with biological units and documentation; (2) the simulation loop is a single readable function with annotated biological identity for each computation; and (3) the architecture exposes explicit cortical-input and thalamic-output interfaces for integration with additional brain-inspired modules.

We note that BioMind-BG does not currently outperform CBGTPy on any quantitative benchmark; its advantage is structural clarity rather than computational novelty. Future work will extend the model with biologically motivated features not present in CBGTPy.

6.3 Limitations

We identify the following limitations, each of which represents a direction for future work:

1. **Single-compartment neuron model.** LIF neurons do not capture dendritic computation, which may be important for MSN coincidence detection and up/down state transitions (Wilson & Kawaguchi, 1996).

2. **No explicit hyperdirect pathway.** The Cx→STN projection is not modeled; STN receives only background input. Including this pathway would enable modeling of response inhibition and speed-accuracy tradeoffs.

3. **Uniform dopamine delivery.** Dopamine is delivered as a uniform signal to all striatal neurons rather than through spatially structured release from SNc terminals.

4. **No cholinergic interneurons.** Tonically active neurons (TANs) modulate the timing window for dopamine-dependent plasticity and are omitted.

5. **No prefrontal control.** The model lacks prefrontal cortical circuits that mediate goal-directed behavior and flexible reversal (as demonstrated by the perseveration in Experiment 3).

6. **Qualitative validation only.** Current validation compares firing rates to literature ranges. Quantitative comparison to specific experimental datasets (e.g., reaction time distributions, choice probability curves) is a target for future work.

7. **Forward Euler integration.** The $dt = 0.2$ ms timestep with forward Euler is adequate for the current parameter regime but may require smaller steps or implicit methods for more detailed biophysical extensions.

8. **Multi-choice scaling.** The model requires explicit connection weight normalization for $N > 2$ action channels (Experiment 5).

6.4 Future Directions

BioMind-BG is designed as the first module of a multi-component, brain-inspired architecture. The next component is the thalamus, which will replace the current simplified thalamic population with a model incorporating burst/tonic mode switching, reticular nucleus gating, and matrix/core

projection specificity. Subsequent components include cortical column microcircuits, hippocampal episodic memory, prefrontal working memory, and a global workspace for information integration. Each component will follow the same design principles established here: biological traceability, single-file parameter documentation, and explicit inter-module interfaces.

We believe that interpretability and biological traceability are not merely desirable properties but essential prerequisites for scaling brain-inspired systems beyond single-region models. As architectures grow to incorporate multiple interacting brain regions, the ability to trace any emergent behavior back to identified biological mechanisms becomes critical for debugging, validation, and principled extension. BioMind-BG establishes this foundation.

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Code and Reproducibility

BioMind-BG is open-source under the MIT License. All code, documentation, and experimental scripts are available at `https://github.com/Rekhii/Biomind`. The model requires only Python 3.8+ and NumPy. All experiments can be reproduced by running `python -m biomind.experiments` with the default random seed (42).

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